MATH 261, Exam 2

Name:

Student ID:

Instructor:

Page Number	Total Points	Points Achieved
3	12	
4	20	
5	20	
6	15	
7	15	
8	18	
Total	100	

Exam Policies

- NO calculators, notecards or electronic devices are allowed during the test. It is your responsibility to make sure you do not have these items on your person. Simply having prohibited materials in an easily accessible location will be considered cheating, regardless of whether you are caught using them.

- Complete work must be shown for problems to receive full credit. Solutions with correct methodology but incomplete final answers may receive partial credit. However, solution containing both correct and incorrect work may result in reduced or no credit. Credit is awarded for accurate work and may be deducted for incorrect statements or calculations.

- Please seek clarification during the examination period if any question or instruction is unclear. Requests for special consideration based on misunderstood questions will not be accommodated after the examination has concluded.

- Valid identification must be presented at exam submission and remain accessible during testing. Ensure your exam includes the correct section and instructor information. You must sign the provided verification list upon submission.

Sections

Ignacio Rojas	001	08:00-08:50 AM	Engineering E105
Iuliana Oprea	002	11:00-11:50 AM	Weber 202
Sam Scheuerman	003	10:00-10:50 AM	Engineering E205
Jerett Cherry	005	4:00-4:50 PM	Engineering E105
Francisco De Jesus Pagan	006	3:00-3:50 PM	Engineering E105
Baohua Chen	007	2:00-2:50 PM	TILT 221
Fernando Herrera	008	11:00-11:50 AM	Engineering 120
Jess Hagman	010	9:00-9:50 AM	Physiology 103

1. (12 points) Evaluate the following integral:

$$\int_0^2 \int_{x/2}^1 \cos(y^2)\ dy\ dx$$

2. (20 points)

Suppose a solid, which is above the cone $z = \sqrt{x^2 + y^2}$ and below the plane $z = 2$, and has a density given by $\delta(x, y, z) = x^2 + y^2$. Set up, but do not solve, a triple integral that represents the **mass** of this solid.

(a) In Cylindrical coordinates.

(b) In Spherical coordinates.

3. (20 points)

Suppose a solid lies within the sphere $x^2 + y^2 + z^2 = 9$, below the plane $z = 2$, and above the plane $z = 0$. Set up a triple integral in an appropriate coordinate system to represent the volume of the solid, and evaluate it.

4. (15 points)

Suppose a wire runs along a curve that is parameterized by $\boldsymbol{r}(t) = \langle t^2,\ 2t,\ e^t \rangle$ for $0 \leq t \leq 1$, and has density $\delta(x, y, z) = x + z$. Set up, but do not solve, a line integral that represents the **mass** of the wire.

5. (15 points)

Compute the work done by the force field $\boldsymbol{F}(x, y) = -y\,\boldsymbol{i} + x\,\boldsymbol{j}$ on a particle moving along the circle $x^2 + y^2 = 4$ once, in a counterclockwise direction.

6. (18 points)

Consider a vector field $\boldsymbol{F} = \langle 2xy + e^x,\ x^2 + \cos(y) \rangle$ and a curve C representing the portion of an ellipse from (2, 0) to (0, 3).

(a) Determine whether **F** is conservative.

(b) If **F** is conservative, find a potential function f. Then use the Fundamental Theorem of Line Integrals to evaluate $\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds$.